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M 206 CIRCULATING WATER PUMPS

1. SCOPE OF WORK

The Bidder shall supply the following in accordance with **Part 4, Subsection 1.2 “Summary of Work”**, including but not limited to.

Two (2) Circulating Water Pumps with a capacity of 50% each per Block, complete with electric motors and all accessories.

Except where otherwise indicated all information in the specification relates to one pump. All pumps supplied shall be identical in every respect.

Each circulating water pump shall include, but not be limited to the following:

- 1) Circulating Water Pump and Motor
- 2) Adjustable shaft coupling
- 3) Thrust bearing
- 4) Lubricating System
- 5) Gland sealing system
- 6) Integral cooling water system
- 7) Base plate
- 8) Foundation, foundation bolts and sleeves
- 9) Instrumentation and control equipment
- 10) Valves, piping, cables, and control boxes
- 11) Electrical Equipment

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of **Part 4, Subsection 2.5, “Codes and Standards”**.

The codes and standards of the following organizations extracted from **Part 4, Subsection 2.5**, or international equivalent standards are specifically applicable to the design, manufacture and testing of the equipment included in this specification:

ANSI	American National Standards Institute.
ASME	American Society of Mechanical Engineers.
ASTM	American Society of Testing and Materials.
BS	British Standard
DIN	German Institute for Standardization
EN	European Committee for Standardization
HIS	Hydraulic Institute Standard.
IEC	International Electrotechnical Commission.
JIS	Japan Industrial Standard

3. DESIGN AND OPERATING CONDITIONS

Circulating water pumps operating in parallel will supply Circulating Water (Cooling Water) to the Condenser, closed cooling water heat exchangers and chlorination system of the Combined Cycle Power Plant.

The design flow shall be based on the peak combined cycle plant operation plus 10% flow margin.

The design of the pumps shall allow the Combined Cycle to be operated at around 60% MCR load in case one of both Cooling water pumps is out of operation.

All pumps shall be designed for continuous parallel operation, and shall have the maximum efficiency at the design flow. In addition the pumps shall be able to operate single (during start-up and following the trip of one pump).

The larger flow of both shall be applied as design flow.

The circulating water pumps shall be installed in the Circulating Water pump house.

Each pump shall operate satisfactorily without difficulty at any point through the full capacity range from zero to run-out condition. Circulating water pump and drive motors shall be capable of taking reverse rotation during emergency.

Bidder shall design total pump head by considering weir design at discharge canal to control constant discharge water level. Pump design and calculation shall be submitted for Owner's review and approval.

Bidder's NPSH-required curve shall include a margin on the minimum suction head required to avoid cavitations (defined as 3 percent total head deviation). The margin shall be 1mWC, and the level of the pump impeller shall be arranged accordingly. Bidder data sheet and outline drawing shall show minimum submergence at run-out and low-low water level at the pumps. Low-low water level at the pumps shall consider high-high differential level across bar racks and travelling screen.

A model test shall be carried out to determine whether or not baffles or other special provisions are needed in the pump suction compartment to ensure proper pump performance. The model test shall also establish the minimum pump submergence for proper operation. The model test shall be carried out by Bidder at Bidder's expense. Tests will be witnessed by the Owner. The model shall be constructed for 1:10 scale. Any special devices including but not limited to flow correcting devices and rearrangement of auxiliary pumps located in the C.W. Intake required as a result of the model test shall be furnished and erected by the Bidder. The model test shall be conducted within 10 month of contract award and results accepted by Owner prior to any intake structure concrete placement of the model, shall include the entire intake structure (to verify cross flow effects) and include any other intake structure pumps (e.g. screen wash, etc).

The Bidder shall conduct and submit result to the owner, a complete hydraulic surge study for the entire C.W. system in order to identify and quantify the existence of water hammer and other transient conditions. The design shall be such as to prevent damage due to these conditions. At the same time the Bidder shall submit the final design hydraulic gradients for the system.

This study shall be submitted with the C.W pipe layout initial drawing submitted.

Pumps shall have non-overloading characteristics, and shall have stable head-capacity characteristics so that the head shall continue to increase with a decreasing capacity until a maximum head is reached at zero capacity. The design head-capacity characteristics shall be as approved by the Owner.

Shut-off head shall not exceed 190% of design head.

Minimum requirements for Circulating Water Pump are as follows:

No.	Description	Requirement
1	Quantity	2 x 50% (Design of each pump shall allow Combine Cycle Plant to operate at around 60% MCR load in case one of both pumps out of operation)
2	Location	Circulating Water Pump House
3	Design flow	As defined in Part 4 Section 3, M-202, clause 3.1 plus closed cooling water flow.
4	Type of operation	Continuous parallel operation, but shall be able to operate single.
5	Vibration ○ Rigid support ○ Flexible support	Within the satisfactory limits as defined by ISO No.3945. Less than 4.6 mm/sec. Less than 7.1 mm/sec.
6	Noise	Shall comply with Part 4 Subsection 2.9 "General Noise Criteria".

4. DETAILED REQUIREMENTS

4.1 Pump General Constructions

The pump shall be of the direct driven, constant speed, vertical spindle, suspended bowl, single stage, mixed flow type for indoor arrangement.

The thrust bearing shall be provided in the driving motor and shall be of the Michell type of ample capacity to carry the weight of all rotating parts plus hydraulic thrust and shall be readily accessible for inspection. The pump shaft bearings shall self-lubricating type (PTFE and Rubber bearing).

The pump seal shall be of the packed gland type with split gland. Any water supply to the seal shall be filtered water taken from the pump discharge. Duplex filtering devices with differential indication and alarms shall be provided.

Each pump and motor drive shall be capable of starting against a closed discharge valve and shall be capable of operating between zero flow and run-out flow without overloading the motor drive. The pump shall be capable of running under severely throttled discharge conditions for purposes of filling and priming the C.W. System. Pump motors shall be in accordance with **Part 4, Section 3, E-211** for Medium Voltage Motor.

Lifting lugs or eyes shall be provided as required to facilitate maintenance handling.

All parts of the pumps and motors shall be designed so that no harm will result if the direction of rotation of pumps is temporarily reversed due to head of water in the discharge pipe. The Bidder shall provide interlocks to ensure that the pumps will not be started under reverse rotation.

Circulating water pumps and auxiliary piping and equipment shall be arranged to allow access for maintenance.

4.2 Discharge Valves

Valves shall be located in the discharge line of each pump. Pump discharge valves shall automatically start to open when the pump motor drive is energized and shall automatically close when pump motor receives a stop signal and prior to stopping the pumps. The pump motor and its associated discharge valve will be energized at the same time. A confirmation of valve operating at 25% open will be received. If confirmation of minimum 25% open is not received in a predetermined time, the pump motor will be tripped. Bidder shall perform the necessary transient analyses to determine the discharge valve opening and closing times which any adverse effects on the system, including any necessary valve operation time delays interlocks shall

be provided to prevent opening of any discharge valve if the outlet valves on the associated condenser are closed.

4.3 Materials

Bidder shall be responsible for providing pumps of design and materials best suited for the specified service conditions. If in the opinion of the Bidder specified materials are not adequate, Bidder shall, in his Bid Proposal, recommend those materials which he considers most suitable for the service.

Materials of construction:

- Casing..... ASTM A240 TYPE 316L or equivalent
- Impeller and impeller shroud Duplex
- Bearing (line) Cutless rubber, backed with Duplex
- Shaft Stainless Steel 316
- Inlet bell Stainless Steel 316L or higher
- Shaft sleeves and wearing rings ... Stainless Steel 316L or higher
- Inner shaft tube Stainless Steel 316 or higher
- Packing glands Stainless Steel 316L or higher

4.4 Painting

Before shipment, all parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter and all exterior ferrous surfaces

shall be painted in accordance to **Part 4 Section 3, S-420 and S 430.**

All exposed machined surfaces shall be coated with a suitable anti-rust compound before shipment.

5. PERFORMANCE, TESTING

5.1 Site Test

After equipment is operational, the Bidder shall run a test to ascertain that the equipment is performing properly in every respect. Any deficiency brought out by this test shall be promptly corrected by the Bidder.

5.2 Shop Test

See **Part 4, Section 3, M-014, “Inspection and Testing”**.

6. QUALITY ASSURANCE / QUALITY CONTROL (QA / QC)

6.1 General

Quality control shall conform to the requirements of **Part 4, Subsection 2.7**.

6.2 Procedure

The quality control procedures, shall include but not be limited to the following:

- Nondestructive testing.
- Packaging and shipping.
- Shop test procedures
- Cleaning and painting.

6.3 Documentation

The quality control documentation shall be as specified in **Part 4, Subsection 2.7**.

6.4 Forgings

Generally refer to **Part 4, Section 3, M-014, “Inspection and testing”**.

The shaft forgings and shaft couplings shall be ultrasonically tested and inspected in accordance with ASTM A388. The scanning shall be continuous over the whole surface of the shaft, and shaft surface for testing shall be smooth enough to obtain an accurate indication.

The entire surface of the shaft forgings shall also be liquid penetrant inspected per ASTM E165.

All other steel forgings shall be inspected by liquid penetrant inspection at the regions of fillets and change of section prior to and after machining. Any flaws which cannot be removed during final machining will be repaired using a repair procedure approved by the Owner.

Wear rings and Coupling shall be liquid penetrant inspected in accordance with ASTM E165.

6.5 Castings

All suction and fabricated components, bowl and discharge column surfaces shall be examined by magnetic particle inspection in accordance with ASTM E709, with basis for acceptance judged by reference to ASTM E125.

All defects beyond the acceptance limits of the above standard requiring weld repair shall be reported to the Owner, and a repair procedure submitted for approval prior to any repair being carried out. Weld repair areas shall be mapped and submitted to the Owner with the test reports.

Following major repairs, all castings shall be fully annealed. Smaller repairs must be followed by stress relieving. All completed repairs shall be fully radiographed.

All weld preparation areas shall be liquid penetrant inspected in accordance with ASTM E165.

The finish machined impellers shall be liquid penetrant inspected to the maximum practicable extent in accordance with ASTM E165.

6.6 Fabrication

All components machined or fabricated from plate, sheet or bar stock shall meet the material requirements of ASTM or a material specifications approved by the Owner.

All weld repair area shall be dye penetrant or magnetic particle examined prior to repair weld deposit.

All tests shall be accordance with the latest recommended practices of ASTM.

6.7 Hydrostatic Testing

The pumps shall be hydrostatically tested to 1.25 times the pumps shut off head or 1.5 times the pump operating head, whichever is greater.

7. INSTRUMENTATION & CONTROLS

7.1 General

The Bidder shall be responsible for the design, supply and installation of complete instrumentation and control devices. The Bidder shall also be responsible for the design, supply and installation of all local instrumentation and control devices and test instruments integral with the Circulating Water Pumps whether or not they are specified herein.

The Bidder shall provide complete logic and other necessary information describing operation of the pumps and their auxiliaries.

Refer to **Part 4 Section 3, I-001, “Instrumentation and Control”** for the design requirements for the instrumentation and control devices. Refer to **Part 4 Section 3, E-001, “General Electrical Specification”** for the design requirements for the junction boxes, cables and wiring.

7.2 Local Control Panel

The Bidder shall furnish and install a control panel with a degree of IP-65 according to IEC 529 for the pumps. The local control panel will only be used for emergency, maintenance and testing purposes. Detailed requirements for local control panels are given **Part 4 Section 3, I-002**.

The local control shall contain all control and monitoring devices which do not need to be installed at a local measuring location, and all operator interfaces such as indicators, push buttons, lights, control switches and alarms.

All instrument and control devices shall be wired to terminal blocks and piped to terminal points as applicable.

Refer to **Part 4, Section I-003** for the design requirements for the local mounted instrument panels.

7.3 Vibration Measuring Devices

The pump shall be provided with vibration measuring devices. Each pumps shall be provided with two radially mounted sensors.

The maximum permissible vibration shall be within the “satisfactory” limits as defined by the International standard ISO No.3945. For rigid support system, the maximum permissible vibration severity (V_{rms}) shall be less than 4.6 mm/sec. and for flexible support system the value shall be less than 7.1 mm/sec.

8. ELECTRICAL SPECIFICATION

8.1 General

The Bidder shall furnish and install all electrical equipment and accessories required for the circulating water pumps.

For general electrical requirements, see **Part 4, Section 3 E-001**.

8.2 Motors

For detailed electrical requirements, see **Part 4, Section 3 E-211** for Medium Voltage Motor.

8.3 Cathodic Protection

For detailed electrical requirements, see **Part 4, Section 3 E-260** C.W. System Cathodic Protection.

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